# MSG No. 222.100 $Pn'\bar{3}'n$ [ Type III, cubic ]

Table 1: Wyckoff site: 2a, site symmetry: 4'32'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,2,3,4,5,6,7,8,9,10,11,12,25,26,27,28,29,30,31,32,33,34,35,36] |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [13,14,15,16,17,18,19,20,21,22,23,24,37,38,39,40,41,42,43,44,45,46,47,48] |

Table 2: Wyckoff site: 6b, site symmetry: 4'2.2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,2,3,4,25,26,33,34] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [5,8,10,12,29,30,31,32] |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [6,7,9,11,27,28,35,36] |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [13,14,21,22,37,38,39,40] |
| 5 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [15,16,23,24,42,43,45,47] |
| 6 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [17,18,19,20,41,44,46,48] |

Table 3: Wyckoff site: 8c, site symmetry: .-3'.

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | [1,5,6,37,41,42] |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [2,10,11,38,46,47] |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [3,7,12,39,43,48] |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [4,8,9,40,44,45] |
| 5 | $\left[0, \frac{1}{2}, 0\right]$ | [13,18,23,25,30,35] |
| 6 | $\left[0, 0, \frac{1}{2}\right]$ | [14,15,19,26,27,31] |
| 7 | $\left[\frac{1}{2}, 0, 0\right]$ | [16,17,21,28,29,33] |
| 8 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [20,22,24,32,34,36] |

Table 4: Wyckoff site: 12d, site symmetry: -4..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | [1,2,13,14] |
| 2 | $\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$ | [3,4,21,22] |
| 3 | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | [5,12,17,19] |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | [6,9,16,23] |
| 5 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | [7,11,15,24] |
| 6 | $\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$ | [8,10,18,20] |
| 7 | $\left[0, \frac{1}{4}, \frac{3}{4}\right]$ | [25,26,37,38] |
| 8 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | [27,36,43,47] |
| 9 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | [28,35,42,45] |
| 10 | $\left[\frac{3}{4}, 0, \frac{1}{4}\right]$ | [29,31,41,48] |
| 11 | $\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$ | [30,32,44,46] |
| 12 | $\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$ | [33,34,39,40] |

Table 5: Wyckoff site: `12e`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | `[1,2,25,26]` |
| 2 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | `[3,4,33,34]` |
| 3 | $\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$ | `[5,12,29,31]` |
| 4 | $\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$ | `[6,9,28,35]` |
| 5 | $\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$ | `[7,11,27,36]` |
| 6 | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | `[8,10,30,32]` |
| 7 | $\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | `[13,14,37,38]` |
| 8 | $\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$ | `[15,24,43,47]` |
| 9 | $\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$ | `[16,23,42,45]` |
| 10 | $\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$ | `[17,19,41,48]` |
| 11 | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | `[18,20,44,46]` |
| 12 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$ | `[21,22,39,40]` |

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ x]$ | `[1,5,6]` |
| 2 | $\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[2,10,11]` |
| 3 | $\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$ | `[3,7,12]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$ | `[4,8,9]` |
| 5 | $\left[-x,\ x+\frac{1}{2},\ -x\right]$ | `[13,18,23]` |
| 6 | $\left[-x,\ -x,\ x+\frac{1}{2}\right]$ | `[14,15,19]` |
| 7 | $\left[x+\frac{1}{2},\ -x,\ -x\right]$ | `[16,17,21]` |
| 8 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[20,22,24]` |
| 9 | $\left[x,\ \frac{1}{2}-x,\ x\right]$ | `[25,30,35]` |
| 10 | $\left[x,\ x,\ \frac{1}{2}-x\right]$ | `[26,27,31]` |
| 11 | $\left[\frac{1}{2}-x,\ x,\ x\right]$ | `[28,29,33]` |
| 12 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[32,34,36]` |
| 13 | $[-x,\ -x,\ -x]$ | `[37,41,42]` |
| 14 | $\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[38,46,47]` |
| 15 | $\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$ | `[39,43,48]` |
| 16 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$ | `[40,44,45]` |

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | `[1,2]` |
| 2 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | `[3,4]` |
| 3 | $\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$ | `[5,12]` |
| 4 | $\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$ | `[6,9]` |
| 5 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$ | `[7,11]` |
| 6 | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | `[8,10]` |
| 7 | $\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | `[13,14]` |
| 8 | $\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$ | `[15,24]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 9 | $\left[\frac{3}{4}, \frac{1}{4}, -x\right]$ | [16,23] |
| 10 | $\left[\frac{1}{4}, -x, \frac{3}{4}\right]$ | [17,19] |
| 11 | $\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$ | [18,20] |
| 12 | $\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$ | [21,22] |
| 13 | $\left[x, \frac{1}{4}, \frac{3}{4}\right]$ | [25,26] |
| 14 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$ | [27,36] |
| 15 | $\left[\frac{1}{4}, \frac{3}{4}, x\right]$ | [28,35] |
| 16 | $\left[\frac{3}{4}, x, \frac{1}{4}\right]$ | [29,31] |
| 17 | $\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$ | [30,32] |
| 18 | $\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$ | [33,34] |
| 19 | $\left[-x, \frac{1}{4}, \frac{3}{4}\right]$ | [37,38] |
| 20 | $\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$ | [39,40] |
| 21 | $\left[\frac{3}{4}, -x, \frac{1}{4}\right]$ | [41,48] |
| 22 | $\left[\frac{1}{4}, \frac{3}{4}, -x\right]$ | [42,45] |
| 23 | $\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$ | [43,47] |
| 24 | $\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$ | [44,46] |

Table 8: Wyckoff site: 24h, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, y, y\right]$ | [1,33] |
| 2 | $\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$ | [2,34] |
| 3 | $\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$ | [3,26] |
| 4 | $\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$ | [4,25] |
| 5 | $\left[y, \frac{1}{4}, y\right]$ | [5,30] |
| 6 | $\left[y, y, \frac{1}{4}\right]$ | [6,27] |
| 7 | $\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$ | [7,28] |
| 8 | $\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$ | [8,29] |
| 9 | $\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$ | [9,36] |
| 10 | $\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$ | [10,31] |
| 11 | $\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$ | [11,35] |
| 12 | $\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$ | [12,32] |
| 13 | $\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$ | [13,40] |
| 14 | $\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$ | [14,39] |
| 15 | $\left[-y, -y, \frac{3}{4}\right]$ | [15,42] |
| 16 | $\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$ | [16,43] |
| 17 | $\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$ | [17,44] |
| 18 | $\left[-y, \frac{3}{4}, -y\right]$ | [18,41] |
| 19 | $\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$ | [19,46] |
| 20 | $\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$ | [20,48] |
| 21 | $\left[\frac{3}{4}, -y, -y\right]$ | [21,37] |
| 22 | $\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$ | [22,38] |
| 23 | $\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$ | [23,47] |
| 24 | $\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$ | [24,45] |

Table 9: Wyckoff site: 48i, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [2] |
| 3 | $\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$ | [3] |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | [4] |
| 5 | $[z,\ x,\ y]$ | [5] |
| 6 | $[y,\ z,\ x]$ | [6] |
| 7 | $\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$ | [7] |
| 8 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$ | [8] |
| 9 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$ | [9] |
| 10 | $\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$ | [10] |
| 11 | $\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$ | [12] |
| 13 | $\left[-x,\ z+\frac{1}{2},\ -y\right]$ | [13] |
| 14 | $\left[-x,\ -z,\ y+\frac{1}{2}\right]$ | [14] |
| 15 | $\left[-z,\ -y,\ x+\frac{1}{2}\right]$ | [15] |
| 16 | $\left[z+\frac{1}{2},\ -y,\ -x\right]$ | [16] |
| 17 | $\left[y+\frac{1}{2},\ -x,\ -z\right]$ | [17] |
| 18 | $\left[-y,\ x+\frac{1}{2},\ -z\right]$ | [18] |
| 19 | $\left[-y,\ -x,\ z+\frac{1}{2}\right]$ | [19] |
| 20 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [20] |
| 21 | $\left[x+\frac{1}{2},\ -z,\ -y\right]$ | [21] |
| 22 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [22] |
| 23 | $\left[-z,\ y+\frac{1}{2},\ -x\right]$ | [23] |
| 24 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [24] |
| 25 | $\left[x,\ \frac{1}{2}-z,\ y\right]$ | [25] |
| 26 | $\left[x,\ z,\ \frac{1}{2}-y\right]$ | [26] |
| 27 | $\left[z,\ y,\ \frac{1}{2}-x\right]$ | [27] |
| 28 | $\left[\frac{1}{2}-z,\ y,\ x\right]$ | [28] |
| 29 | $\left[\frac{1}{2}-y,\ x,\ z\right]$ | [29] |
| 30 | $\left[y,\ \frac{1}{2}-x,\ z\right]$ | [30] |
| 31 | $\left[y,\ x,\ \frac{1}{2}-z\right]$ | [31] |
| 32 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [32] |
| 33 | $\left[\frac{1}{2}-x,\ z,\ y\right]$ | [33] |
| 34 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [34] |
| 35 | $\left[z,\ \frac{1}{2}-y,\ x\right]$ | [35] |
| 36 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | [36] |
| 37 | $[-x,\ -y,\ -z]$ | [37] |
| 38 | $\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [38] |
| 39 | $\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | [39] |
| 40 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | [40] |
| 41 | $[-z,\ -x,\ -y]$ | [41] |
| 42 | $[-y,\ -z,\ -x]$ | [42] |
| 43 | $\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$ | [43] |
| 44 | $\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$ | [44] |
| 45 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$ | [45] |
| 46 | $\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [46] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 47 | $\left[-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$ | [47] |
| 48 | $\left[z+\frac{1}{2},\, -x,\, y+\frac{1}{2}\right]$ | [48] |